Gaussian Process

A probabilistic approach to learning unknown functions

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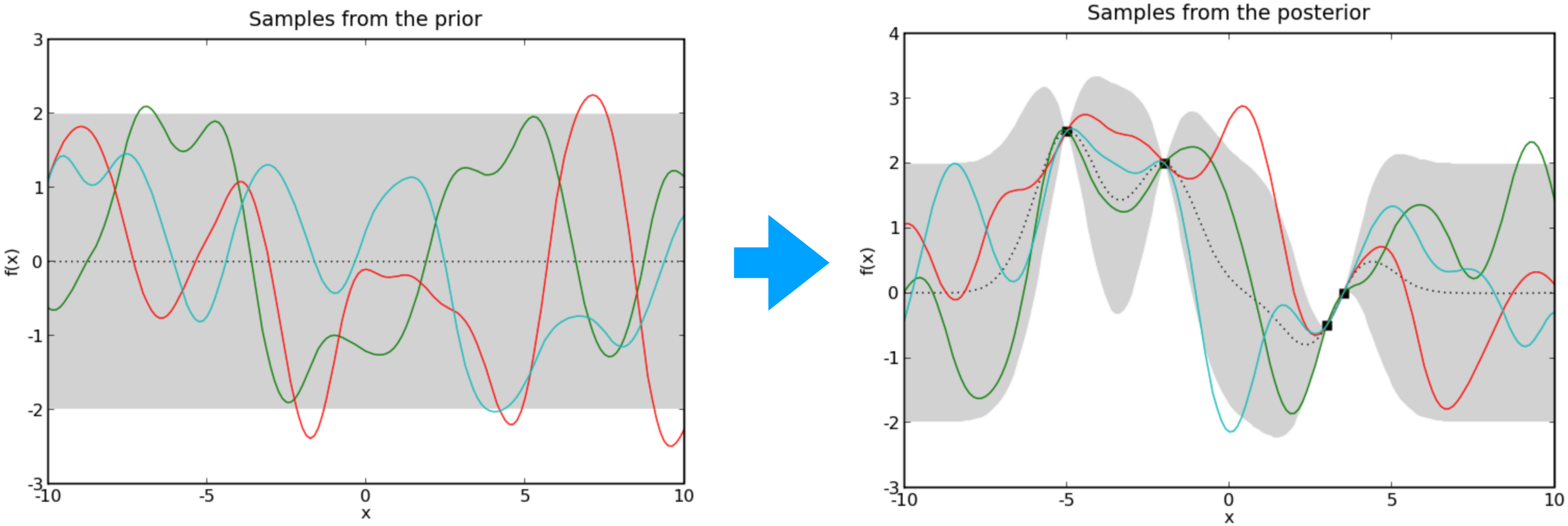
What you can do with Gaussian Process Regression

- Providing plausible curves (or surfaces) with plausible uncertainty regions when the functional form of the data is unknown -> useful for interpolation



What we learn…

1. What is a Gaussian Process?
2. What does a kernel describe?
3. How do observations transform a GP prior into a posterior?
4. How do we choose and train a GP model?

From linear regression to a Gaussian Process

Traditional Regression Models

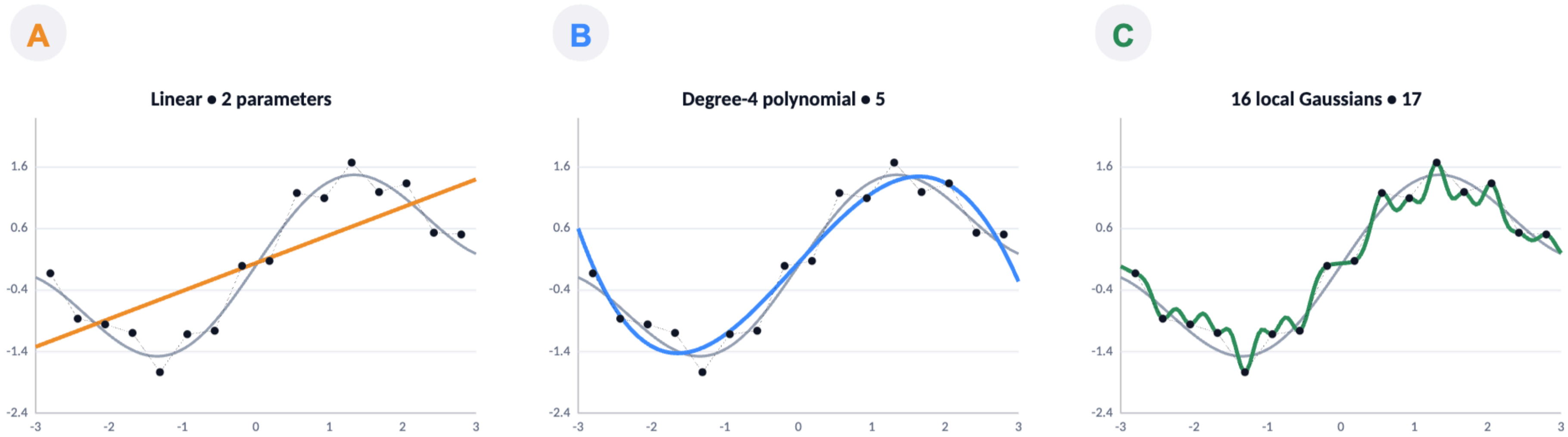
$$y_i = f(x_i; \theta) + \varepsilon_i$$

- Choose a functional form f first
- the number of parameters θ is finite
- errors are often assumed independent
- the assumed functional form may be wrong

Linear Regression

More flexible linear model with basis functions

$$\boxed{\mathbf{f} = \Sigma w_j \phi_j(x) = \phi(x)^T \mathbf{w}}$$



Few bases → underfit (high bias) | many bases → possible overfit (high variance)

Bayesian linear regression gives a distribution over functions

- "For any set of N input points, $(x_1, ..., x_N)$, if the joint distribution p(**f**) of the corresponding outputs **f**=$(f(x_1), ..., f(x_n))$ is a multivariate Gaussian distribution, then the relationship between x and f is described by a **Gaussian process**."

$$\mathbf{f} = \mathbf{\Phi w} \quad \text{where} \quad \mathbf{w} \sim \mathcal{N}(0, \Sigma)$$

$$\Downarrow$$

$$\mathbf{f} \sim \mathcal{N}(0, \mathbf{\Phi} \Sigma \mathbf{\Phi}^T)$$

A probability distribution over coefficients induces a probability distribution over functions

Kernel Trick

$$\mathbf{K} = \mathbf{\Phi}\Sigma_w\mathbf{\Phi}^T \Rightarrow K_{nm} = \Sigma\phi^T(x_n)\phi(x_m)$$

We can define a kernel function such that

$$k(x_n, x_m) = \phi(x_n)^T\Sigma_w\phi(x_m)$$

Note that you don't need to define $\phi(x)$ explicitly. We can just define a kernel function.

Kernel function does not need to be described by the finite number of basis functions (i.e., the dimension of the corresponding basis functions can be infinity)

Kernel matrix needs to be symmetric and positive semi-definite.

Gaussian Process prior and kernels

Definition of a Gaussian Process

- The joint distribution over any finite sample f(x) drawn from a GP is a multivariate Gaussian:

$$\boxed{f(x) \sim \mathcal{GP}(m(x), k(x, x'))}$$

- For any $x_1, \ldots, x_n$,

$$\begin{pmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{pmatrix} \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

where $\mathbf{m}$ is a mean function and $\mathbf{K}$ is a kernel function

What Gaussian Process Regression does

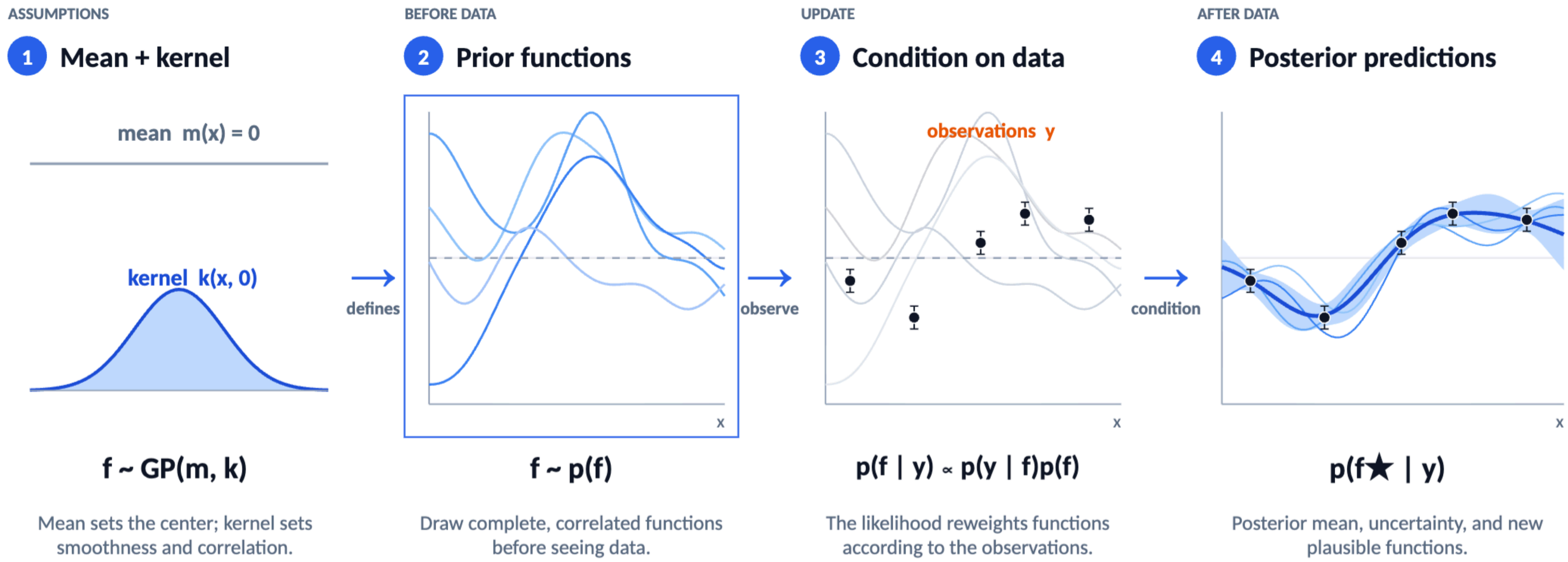
$$\mathrm{f(x)} \sim \mathrm{GP(m,\ k)}$$

Mean function

Covariance/
Kernel function

- Specify which functions are plausible through a mean function and a kernel
- Use the observations to update this prior distribution
- Return a posterior distribution at every prediction point.

What Gaussian Process Regression does



The kernel shapes how function values move together; conditioning reweights those functions using the data.

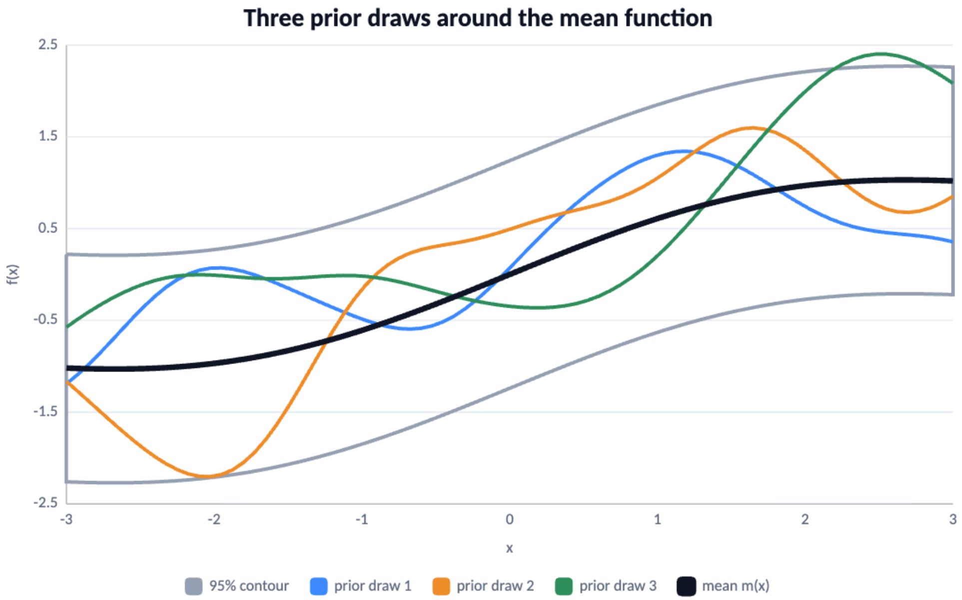
Mean function and covariance function

Mean function

$$m(x) = E[f(x)]$$

Covariance kernel

$$k(x, x') = \mathrm{Cov}[f(x), f(x')]$$

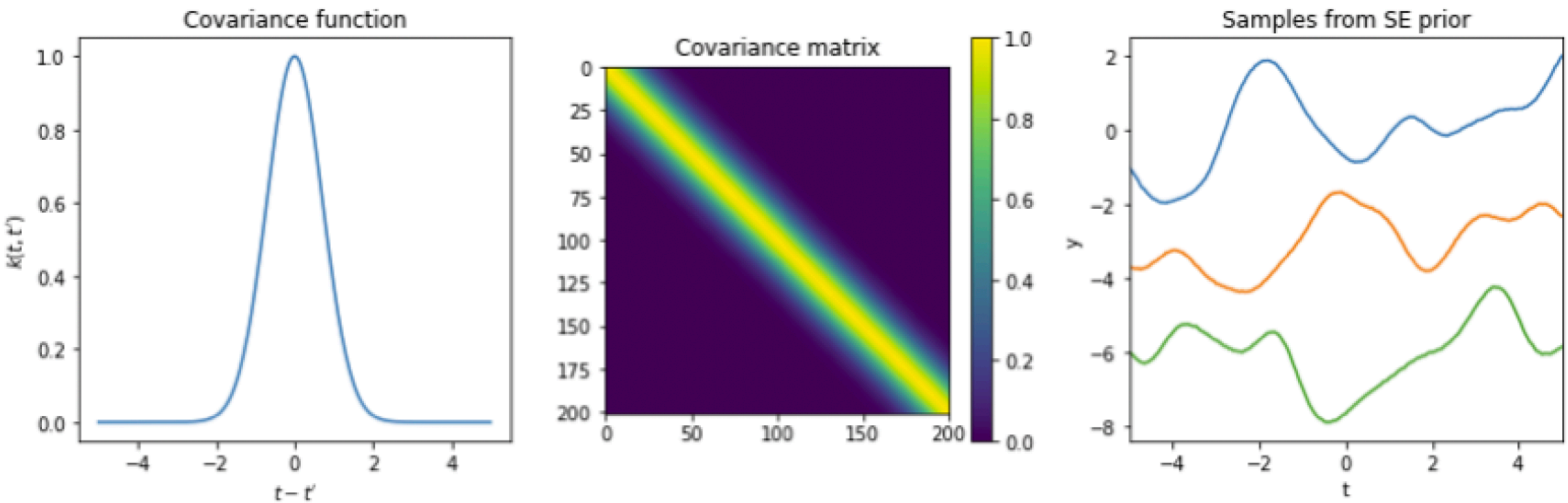


Mean function m(x) sets the central trend.

Kernel k(x, x′) sets spread and smoothness.

Together: f ~ GP(m, k) defines the prior.

From a kernel function to a covariance matrix



$k(x, x')$ as a function of separation

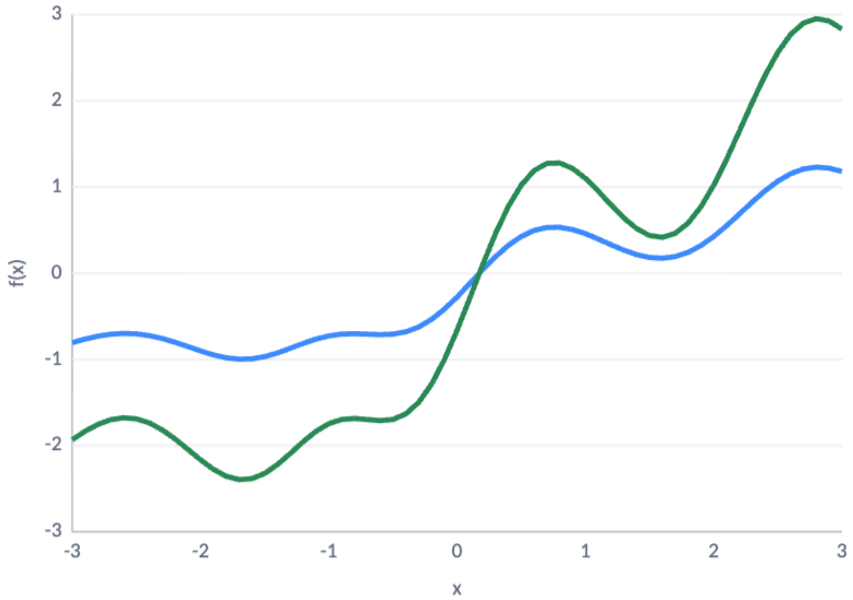
$K_{nm} = k(x_n, x_m)$ as a covariance matrix

$\mathbf{f} \sim \mathcal{N}(\boldsymbol{m}, \mathbf{K})$:Prior function samples, vertically offset for clarity

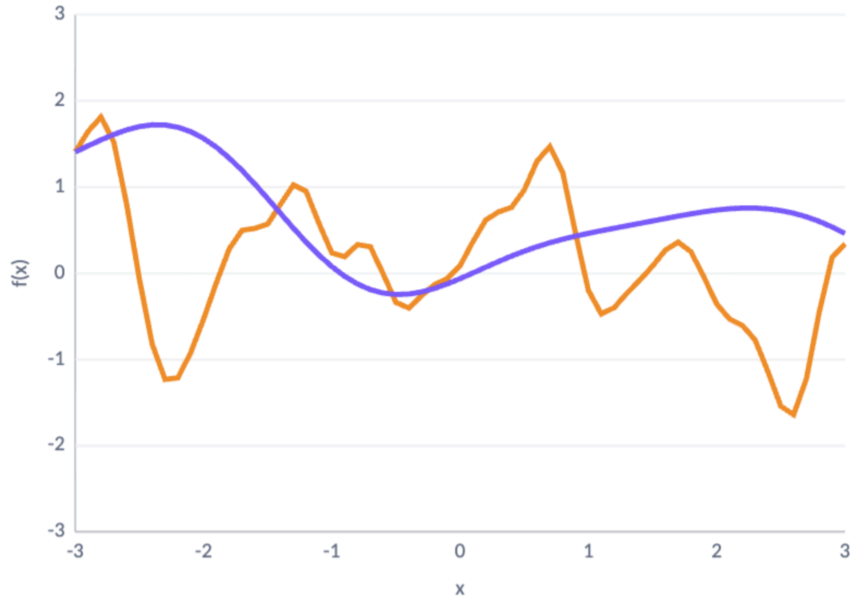
Hyperparameters determine prior assumptions

$$k(\boldsymbol{x}_i, \boldsymbol{x}_j) = \theta_1 \exp\left\{-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2}{\theta_2}\right\}$$

θ_1: 0.5 (blue) → 1.2 (green); θ_2 = 0.7



θ_2: 0.25 (orange) → 1.2 (purple); θ_1 = 1



kRBF(x,x′) = $\theta_1^2 \exp[-(x-x')^2/(2\theta_2^2)]$ | θ_1: amplitude θ_2: length scale

Gaussian Process Regression

From a latent function to noisy observations

- Observed data y can be described with $f(x)$ and noise ϵ:

$$y = f(x) + \epsilon$$

- Assume Gaussian observational noise:

$$\epsilon \sim \mathcal{N}(0, \mathbf{\Sigma}_\epsilon)$$

- Then, the probability distribution for y is given by

$$p(y \mid x) = \mathcal{N}(\mathbf{m}, \mathbf{K}_f + \mathbf{\Sigma}_\epsilon)$$

More generally, we can describe $\mathbf{K} = \mathbf{K}_f + \mathbf{\Sigma}_\epsilon$ for the correlated noise

Noisy training data and a latent test value are jointly Gaussian

- Let $\mathbf{x} = (x_1, \ldots, x_n)$ be the training inputs and let $x_\star$ be a new test input. Then, we can find $f_\star = f(x_\star)$ as

$$\begin{pmatrix} \mathbf{y} \\ f_\star \end{pmatrix} \sim \mathcal{N}\left[\begin{pmatrix} \mathbf{m} \\ m_\star \end{pmatrix}, \begin{pmatrix} \mathbf{K}_y & \mathbf{k}_\star \\ \mathbf{k}_\star^{\mathrm{T}} & k_{\star\star} \end{pmatrix}\right]$$

- $K_y = \mathbf{K}_f + \boldsymbol{\Sigma}_\epsilon$: covariance among observations
- $\mathbf{k}_\star$: covariance between observations and prediction points
- $k_{\star\star}$: covariance among prediction points.
- $f_\star = f(x_\star)$: unknown latent signal at $x_\star$

K_y includes training noise, but $\mathbf{k}_\star$ and $k_{\star\star}$ do not include future noise

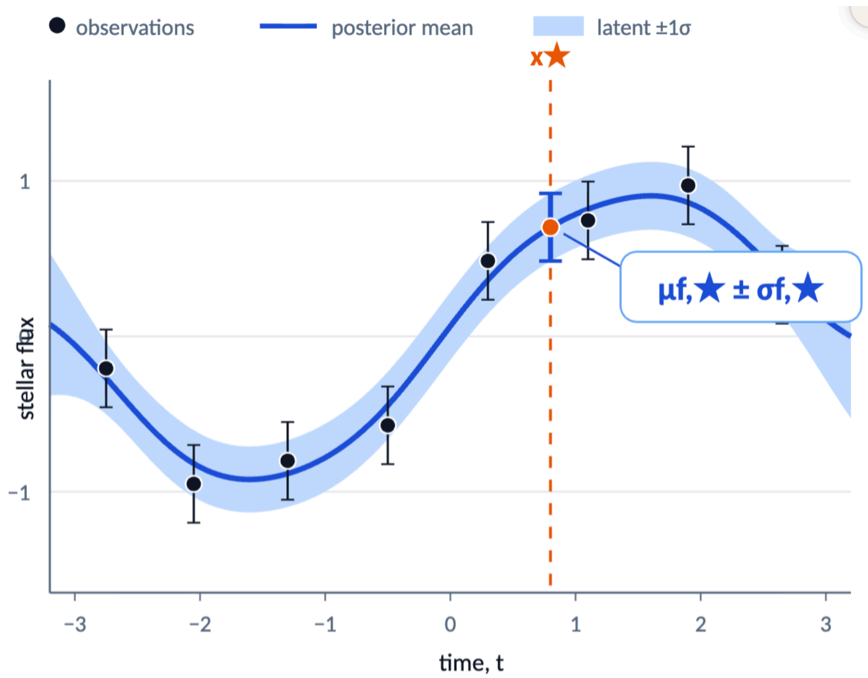
Conditioning predicts the underlying latent function

- Given $\mathscr{D} = \{(x_1, y_1), \ldots, (x_n, y_n)\}$, you get the predictive distribution

$$p(f_\star \,|\, x_\star, \mathscr{D}) = \mathscr{N}(\mu_{f,\star}, \Sigma_{f,\star})$$

where $\mu_{f,\star} = m_\star + \mathbf{k}_\star^T \mathbf{K}_y^{-1}(\mathbf{y} - \mathbf{m})$ and $\Sigma_{f,\star} = k_{\star\star} - \mathbf{k}_\star^T \mathbf{K}_y^{-1} \mathbf{k}_\star$

- These are the predictive mean and variance obtained by conditioning the joint Gaussian distribution.



Posterior uncertainty in the latent physical signal

Posterior mean = prior mean + data correction

$$\boxed{\mu_\star = m_\star + \mathbf{k}_\star^T \mathbf{K}_y^{-1}(\mathbf{y} - \mathbf{m})}$$

- $m_\star$: baseline prediction (can be your model)
  - specified from prior physical knowledge
  - optimized jointly with the kernel hyperparameters
- $\mathbf{k}_\star^T \mathbf{K}_y^{-1}(\mathbf{y} - \mathbf{m})$: correction learned from data
- The posterior mean is a weighted sum of kernel functions centred on the observations

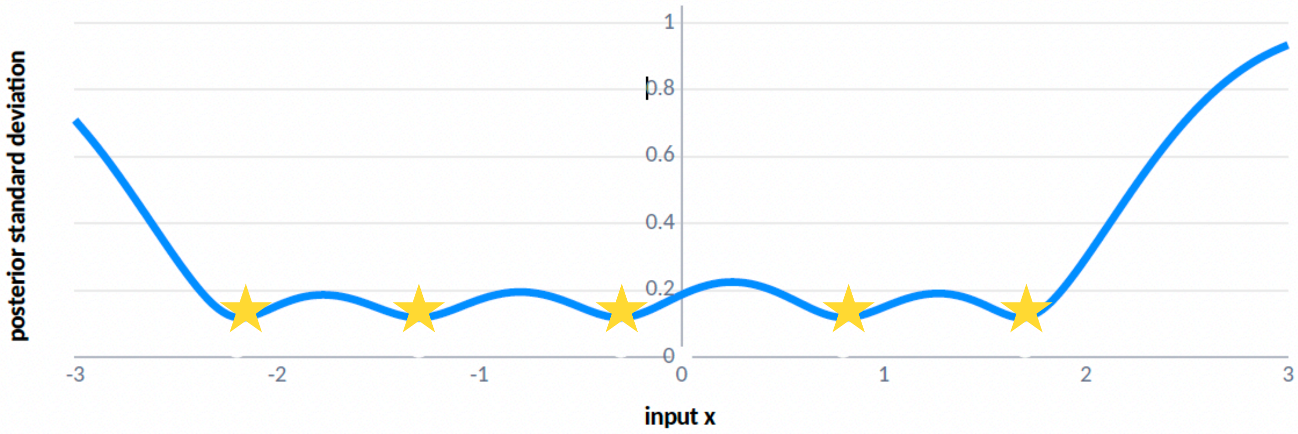
Posterior covariance = prior uncertainty – explained uncertainty

$$\boxed{\boldsymbol{\Sigma}_\star = \mathbf{k}_{\star\star} - \mathbf{k}_\star^T \mathbf{K}_y^{-1} \mathbf{k}_\star}$$

- $\mathbf{k}_{\star\star}$: prior uncertainty
- $\mathbf{k}_\star^T \mathbf{K}_y^{-1} \mathbf{k}_\star$: information provided by the data
- the predictive covariance depends on observation locations, not their measured values
- adding informative data reduces uncertainty

Uncertainty is smallest near informative data

$$\boxed{\Sigma_\star = \mathbf{k}_{\star\star} - \mathbf{k}_\star^T \mathbf{K}_y^{-1} \mathbf{k}_\star}$$



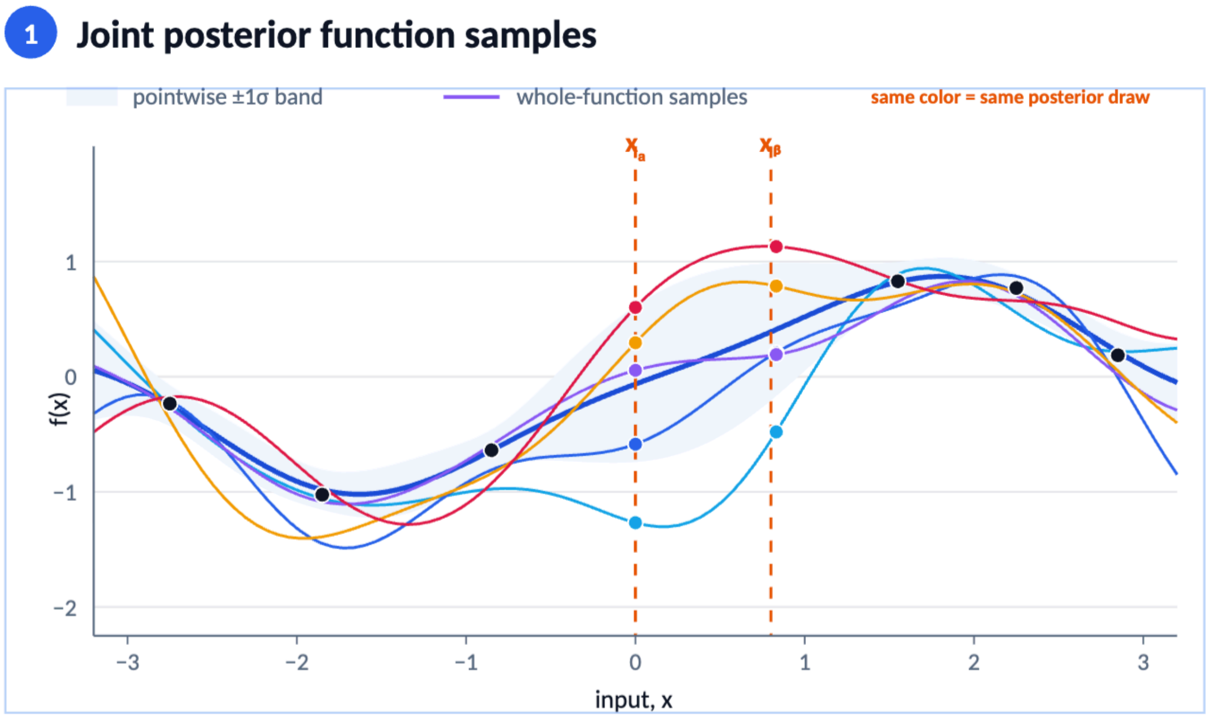
Near informative data

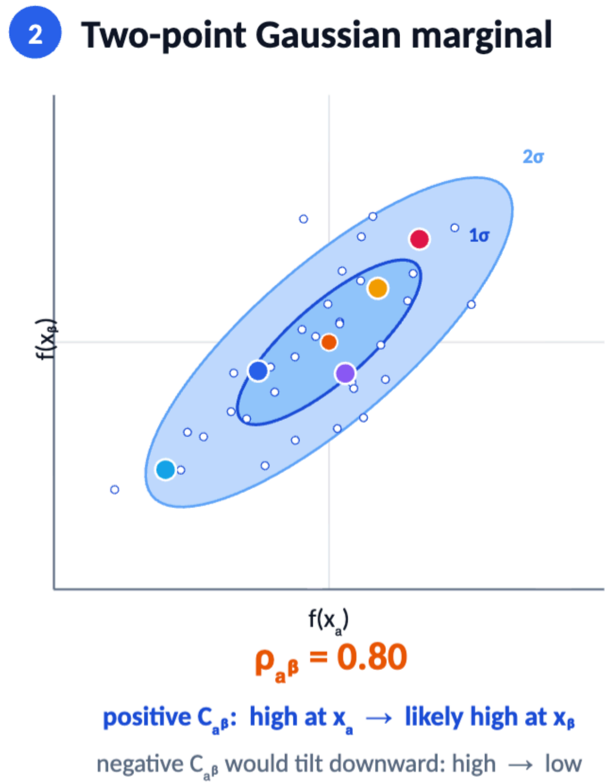
Uncertainty shrinks because correlated variation is constrained.

Far from data

Uncertainty returns toward the prior variance.

Predictions at different inputs are correlated





If one posterior function is high at x_a, the covariance determines whether it is also likely to be high or low at x_β.

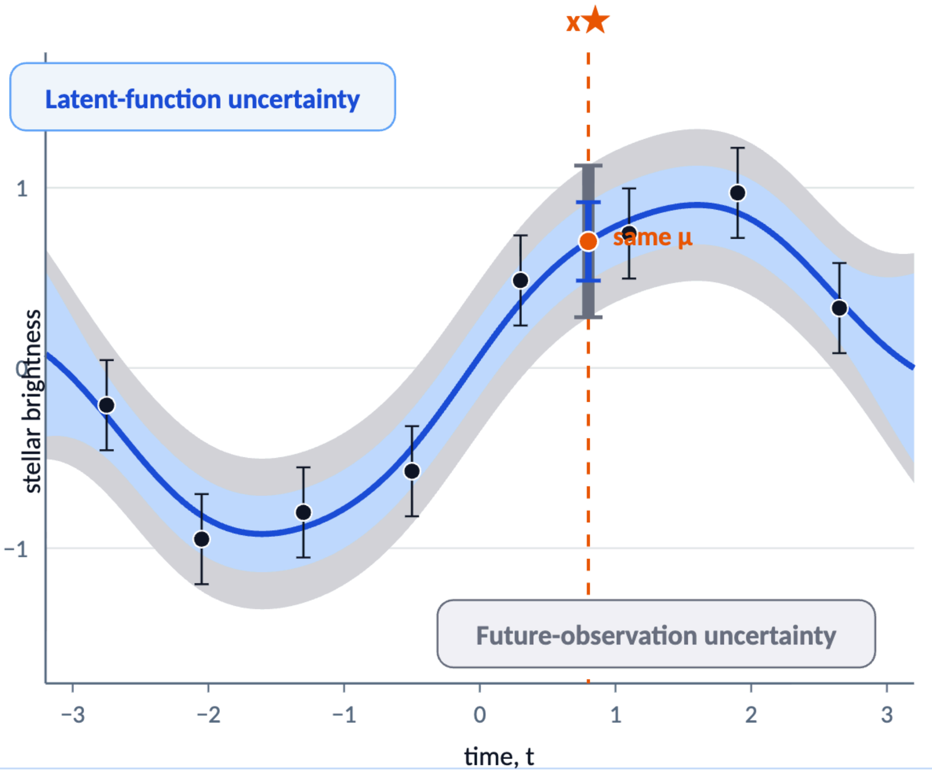
A future observation contains additional measurement noise

- The future measurement contains noise:

$$y_\star = f_\star + \epsilon_\star \text{ with } \epsilon \sim \mathcal{N}(0, \Sigma_\epsilon)$$

- Then,

$$p(y_\star \mid x_\star, \mathcal{D}) = \mathcal{N}(\mu_{f,\star}, \Sigma_{f,\star} + \Sigma_\epsilon)$$



Same mean, but different variance!

Learning and choosing the GP model

Learning hyperparameters with the marginal likelihood

$$p(\mathbf{y} \mid \mathbf{X}, \theta) = \mathcal{N}(\mathbf{y} \mid \mathbf{m}, \mathbf{K}_\theta)$$

$$\log p(\mathbf{y} \mid \mathbf{X}, \eta) = -\frac{N}{2}\log(2\pi) - \frac{1}{2}\log|\mathbf{K}_\mathrm{y}| - \frac{1}{2}(\mathbf{y}-\mathbf{m})^\mathrm{T}\mathbf{K}_\mathrm{y}^{-1}(\mathbf{y}-\mathbf{m})$$

$$\propto -\log|\mathbf{K}_\mathrm{y}| - (\mathbf{y}-\mathbf{m})^\mathrm{T}\mathbf{K}_\mathrm{y}^{-1}(\mathbf{y}-\mathbf{m})$$

- $(\mathbf{y}-\mathbf{m})^T\mathbf{K}_y^{-1}(\mathbf{y}-\mathbf{m})$: data fitting
- $\log|\mathbf{K}_\mathrm{y}|$: penalty for complexity
- η includes all the hyper parameters for mean, kernel, and noise

Learning hyperparameters with the marginal likelihood

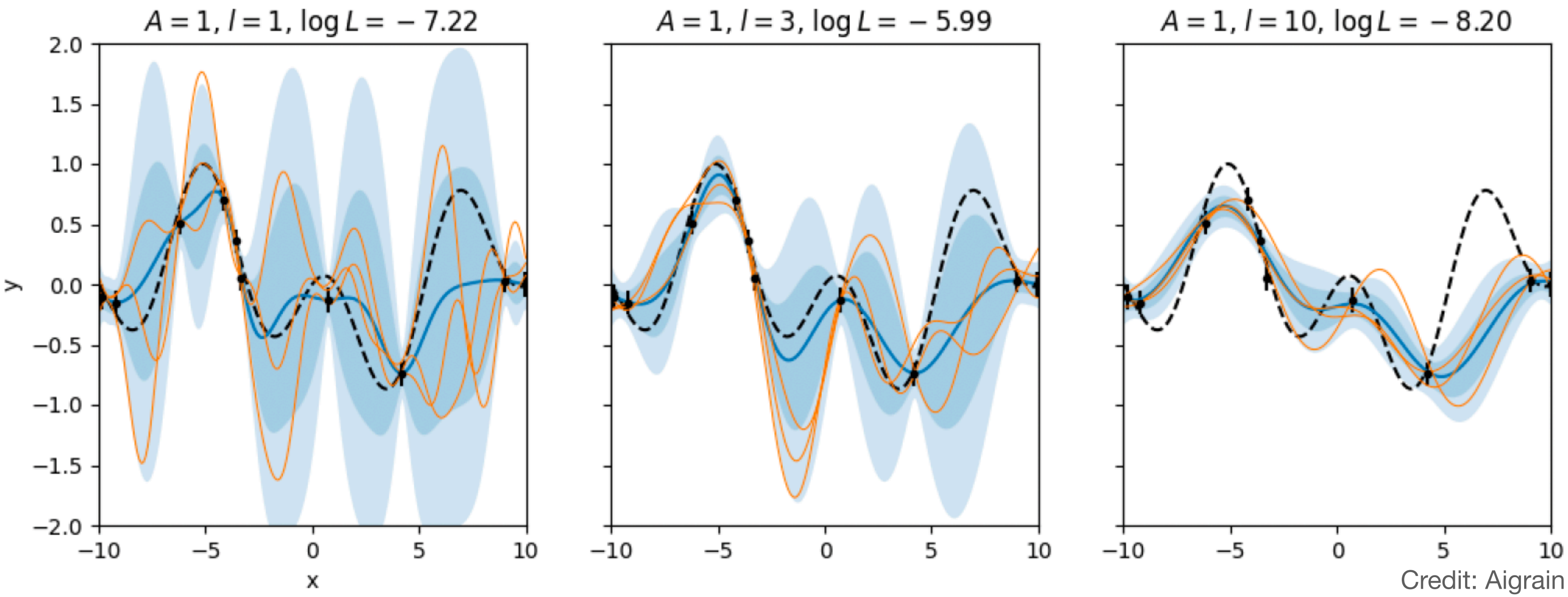
$$\log p(\mathbf{y} \mid \mathbf{X}, \eta) = -\frac{N}{2}\log(2\pi) - \frac{1}{2}\log|\mathbf{K}_{\mathrm{y}}| - \frac{1}{2}(\mathbf{y} - \mathbf{m})^{\mathrm{T}}\mathbf{K}_{\mathrm{y}}^{-1}(\mathbf{y} - \mathbf{m})$$
$$\propto -\log|\mathbf{K}_{\mathrm{y}}| - (\mathbf{y} - \mathbf{m})^{\mathrm{T}}\mathbf{K}_{\mathrm{y}}^{-1}(\mathbf{y} - \mathbf{m})$$

- **Optimization**:select one best-fit hyperparameter set $\hat{\eta} = \arg\max_{\eta} \log \mathrm{p}(\mathbf{y} \mid \eta)$
- **Bayesian Marginalization**: assign hyperpriors, sample $p(\eta \mid \mathbf{y})$, and average predictions over plausible hyperparameter values.

Training GP

- Given a dataset and a choice of mean and covariance function, we can look for the set of hyper-parameters that maximize the likelihood

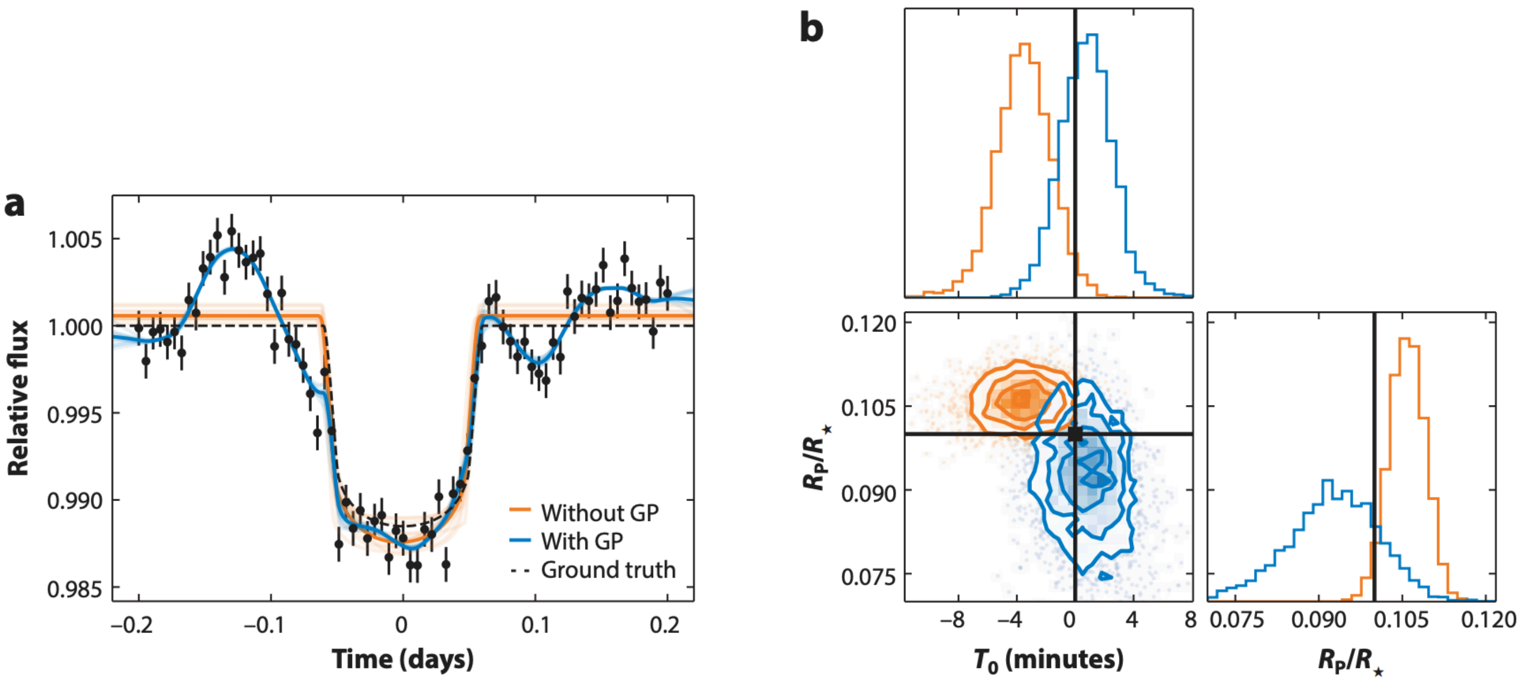
$$k(\boldsymbol{x}_i, \boldsymbol{x}_j, \boldsymbol{\phi}) = A^2 \exp\left(-\frac{\tau^2}{2l^2}\right)$$



Credit: Aigrain

Example: Exoplanet transit with stellar variability

- Fitting simulated observations of a planetary transit with correlated noise with a GP to measure the transit parameter robustly.



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Types of Kernel Functions

Name	Representation[a]
Constant	α^2
Squared exponential[b]	$e^{-(\tau/\lambda)^2/2}$
Exponential[c]	$e^{-\tau/\lambda}$
Matérn-3/2	$\left(1+\sqrt{3}\tau/\lambda\right) e^{-\sqrt{3}\tau/\lambda}$
Matérn-5/2	$\left(1+\sqrt{5}\tau/\lambda+5(\tau/\lambda)^2/3\right) e^{-\sqrt{5}\tau/\lambda}$
Rational quadratic	$\left(1+\frac{\tau^2}{2\gamma\lambda^2}\right)^{-\gamma}$
Cosine	$\cos 2\pi\tau/\lambda$
Sine squared exponential	$\exp\left(-\Gamma \sin^2 \pi\tau/\lambda\right)$
Stochastic harmonic oscillator[d]	$\cos\left(\sqrt{1-\beta^2}\,\frac{\tau}{\lambda}\right)+\frac{\beta}{\sqrt{1-\beta^2}}\sin\left(\sqrt{1-\beta^2}\,\frac{\tau}{\lambda}\right)$

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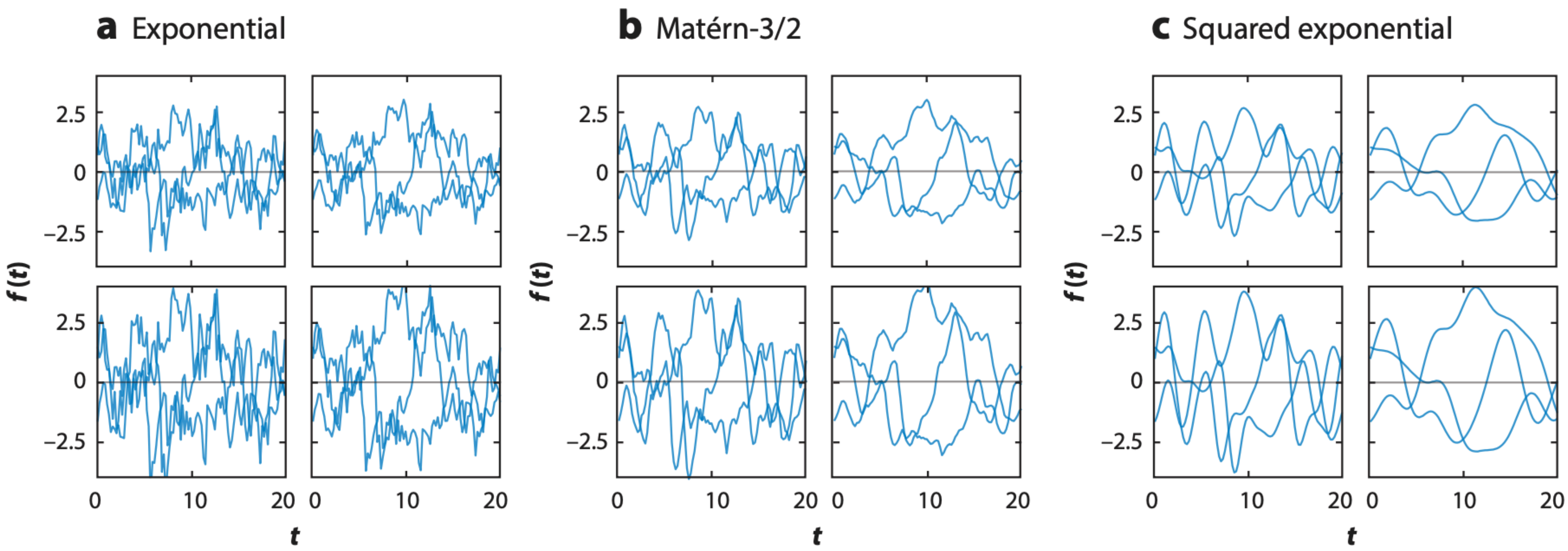
Kernel example

- Different kernels have different smoothness and characteristic input/output scales of the allowed functions change with the hyper-parameters



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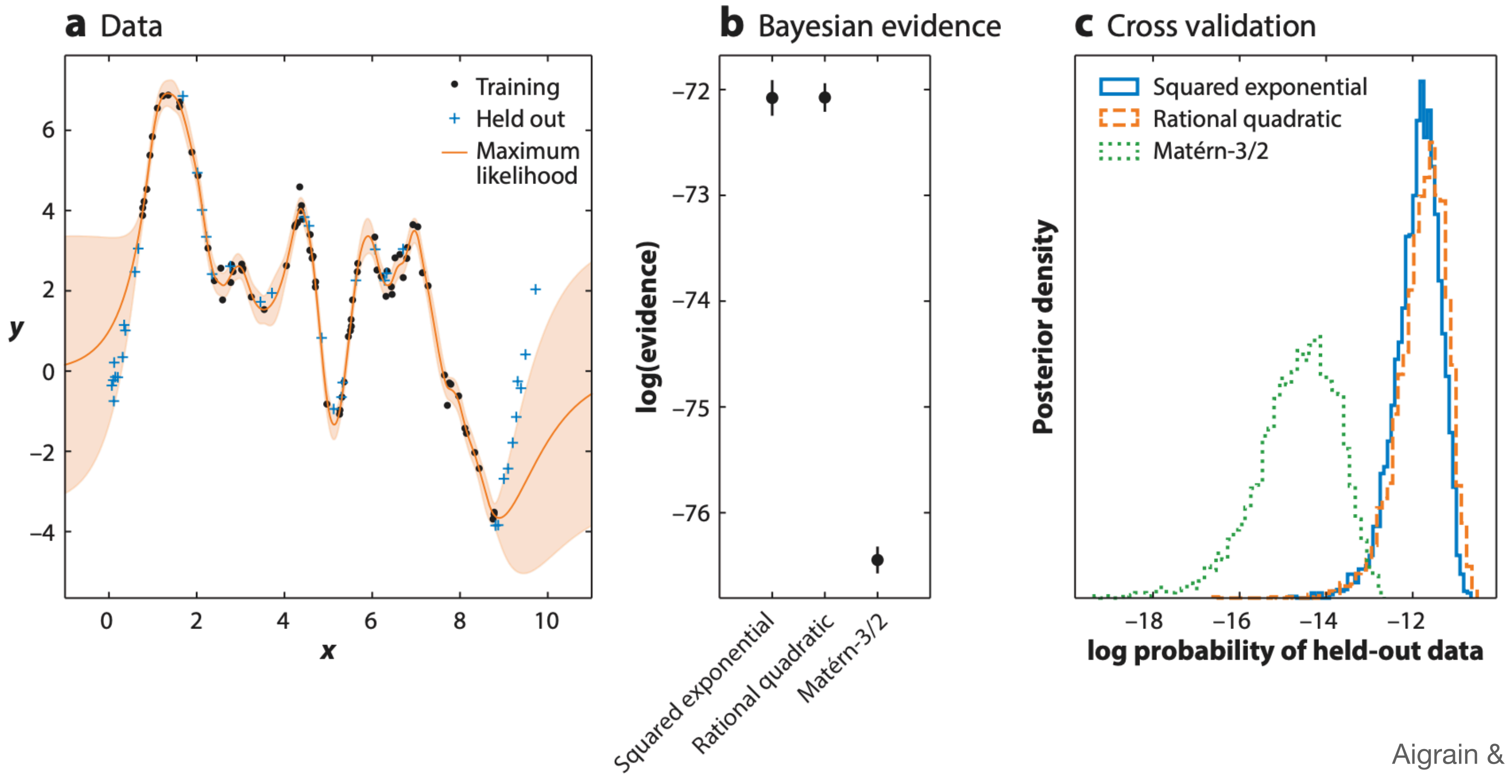
Model Assessments

- A good GP must predict held-out data—not merely fit the training data.



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When should we be cautious with GPs?

- the kernel may be misspecified;
- a flexible GP can absorb the physical signal of interest;
- exact GP inference scales poorly with N;
- non-Gaussian data may require a different likelihood;
- extrapolation is controlled mainly by the prior.

Summary

$$\boxed{f(x) \sim \mathcal{GP}(m(x), k(x, x'))}$$

- A GP is a probability distribution over functions.
- The mean function specifies the baseline.
- The kernel specifies covariance and prior function structure.
- Conditioning on data gives a posterior mean and covariance.
- Kernel and hyperparameter choices are assumptions that must be validated.